Introduction to Software Engineering

Project Proposal & Plan

The student team is required to complete the **Project Proposal & Plan** documentation for the assigned course project, following the attached template.



Software Engineering Department
Faculty of Information and Technology
University of Science

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Project Proposal & Plan

Objectives

This document focus on the following topics:

- ✔ Completing the Project Proposal & Plan document with the following sections:
  - Vision
  - Proposed Solution
  - Project Plan
  - Risk Management
  - Tools & Collaboration Setup
- ✔ Including an AI usage log for this phase
- ✔ Understanding the Project Proposal & Plan document.

1 Member Contribution Assessment

[List every member. Contribution (%) must add up to 100%. Evidence: links to tasks on the board, commits or pull requests.]

ID	Name	Role(s)	Contribution (%)	Evidence
24127248	Nguyễn Trần Phương Thúy	Business Analyst	30%	https://github.com/Trananhkhoa2929/ImpactHub/issues/5 https://github.com/Trananhkhoa2929/ImpactHub/issues/9 https://github.com/Trananhkhoa2929/ImpactHub/issues/10 https://github.com/Trananhkhoa2929/ImpactHub/issues/13 https://github.com/Trananhkhoa2929/ImpactHub/issues/12
24127328	Hồng Gia Bảo	Designer	20%	https://github.com/Trananhkhoa2929/ImpactHub/issues/11 https://github.com/Trananhkhoa2929/ImpactHub/issues/8
24127424	Trần Anh Khoa	Project Manager / Tester	30%	https://github.com/Trananhkhoa2929/ImpactHub/issues/10 https://github.com/Trananhkhoa2929/ImpactHub/issues/12
24127508	Lê Nho Duy Phước	Developer	20%	https://github.com/Trananhkhoa2929/ImpactHub/issues/6 https://github.com/Trananhkhoa2929/ImpactHub/issues/7

2 Vision

2.1 Preliminary Problem Statement

Nowadays, social media platforms are widely used to share stories about people in need and encourage community support. However, information about charitable cases is often scattered across different platforms, making it difficult for users to discover fundraising campaigns and follow their progress. Although social media supports communication and interaction, it generally lacks integrated tools for managing fundraising activities and tracking donations. Meanwhile, crowdfunding platforms mainly focus on fundraising rather than community discussions and interactions.

To address these problems, **ImpactHub** is proposed as a web application that combines a community forum, fundraising campaign management, and donation tracking in a single platform. The system aims to connect people in need with individuals who are willing to provide support while making fundraising activities easier to organize and follow.

ImpactHub supports several types of users, including donors, campaign organizers, and administrators. Users can register, log in, manage their profiles, and access features according to their roles. Through the community forum, they can share stories, upload images, comment, react, share, and search for posts. Campaign organizers can create and manage fundraising campaigns, set donation goals, and monitor fundraising progress. Donors can make simulated donations and view campaign progress through progress bars. Administrators can manage users, moderate forum posts, and approve or reject fundraising campaigns.

Campaign approval is limited to the system's technical workflow and administrator permissions. The system does not determine whether a campaign is legitimate or trustworthy. Official identity verification, background checks, and legal verification are outside the project's scope. In addition, volunteer management and the charity marketplace are considered optional features and are not prioritized over the main functions.

Operating Environment

ImpactHub is designed as a responsive web application accessible through modern web browsers supporting HTML5 on desktop and mobile devices. Users can access the platform through an Internet connection without installing a separate application. The system is expected to operate using a cloud-hosted backend and database to support user accounts, forum content, fundraising campaigns, and donation records.

The proposed deployment environment includes Vercel for the frontend and Render for the backend and database. These platforms are selected for their accessibility and suitability for a student project. The final deployment configuration will be confirmed during development.

Design and Implementation Constraints

The proposed frontend technologies are React with Vite and Tailwind CSS, while the backend is planned to use Node.js and Express with JavaScript. PostgreSQL is proposed for storing user information, forum posts, fundraising campaigns, and donation records. GitHub will be used for source code management, Figma for user interface design, and Postman for testing application programming interfaces (APIs).

The system must support role-based access to ensure that users can perform only the actions permitted by their roles. Donation functionality will operate in a simulated environment, without processing real-money transactions or bank withdrawals. The project also excludes advanced fraud detection and official identity verification.

Due to the limited development time and team resources, implementation will prioritize the five Must-have features identified in the project scope. The team will follow the course's software engineering process and documentation requirements, including requirements specification, system design, testing, and project planning. Detailed functional and non-functional requirements will be defined in PA2.

2.2 Product Position Statement

Element	Content
For	People who want to support charitable causes (donors), campaign organizers, and community participants.
Who	Need a trustworthy, convenient way to discover charitable stories, manage approved fundraising campaigns, discuss community needs, and follow donation progress.
The (product name)	ImpactHub – a community engagement and fundraising web application.
That	Combines community discussion, administrator-reviewed campaigns, simulated donations, and visible fundraising progress in one place.
Unlike	Scattered social-media posts and separate crowdfunding platforms such as Threads, GoFundMe, and Give.asia.
Our product	Integrates a searchable community forum with campaign approval, role-based management, and transparent demo donation tracking in a single responsive website.

2.3 Stakeholders and Users

#	Stakeholder / user	Description	Main interests
1	Project Team	Delivering a functional system within the project scope.	Develops and maintains ImpactHub.
2	Donors	Discovering campaigns, donating, and tracking progress.	Users who support fundraising campaigns.
3	Campaign Organizers	Managing campaigns and monitoring donations.	Users who create and manage campaigns.
4	Administrators	Managing users, approving campaigns, and moderating posts.	Users responsible for system management and moderation.

2.4 Scope and Key Features

ID	Feature	Description	Priority (M/S/C/W)

F01	Account & role management	Users sign up, log in (email + password, JWT), edit their profile, and access features according to their role (Donor, Organizer, Volunteer, Admin).	Must
F02	Community forum	Users create posts with images, comment, react, share a post link and search posts by keyword.	Must
F03	Fundraising campaign management	Campaign organizers can create, edit, and close fundraising campaigns with a title, story, cover image, donation goal, and deadline. Campaigns must be approved by an administrator before becoming publicly visible.	Must
F04	Campaign approval & moderation	Administrators review and approve or reject fundraising campaigns based on platform rules and moderate reported forum posts. This review does not include official identity verification, legal checks, or background investigations.	Must
F05	Donation & progress tracking	Users make simulated donations through a demo wallet; campaign totals and progress bars update right after each donation.	Must
F06	Expense reporting	Organizers post expense reports for a campaign; admins review them and donors can view them for transparency.	Should
F07	Email notifications	The system emails users about verification results, campaign approval and donation receipts.	Should
F08	Volunteer task board	Organizers create simple volunteer tasks; volunteers register and update task status (To do / Doing / Done).	Should
F09	Payment gateway (sandbox)	Donations can go through Stripe or VNPay in test mode instead of the demo wallet.	Could
F10	Charity marketplace	Users list and buy products (simulated payment); a set percentage of each sale goes to a chosen campaign.	Could

F11	Google login	Users can sign in with a Google account.	Could
F12	Native mobile app	A separate Android/iOS application.	Won't
F13	AI fraud detection & auto-moderation	Automatically detecting fake campaigns or harmful posts with AI.	Won't

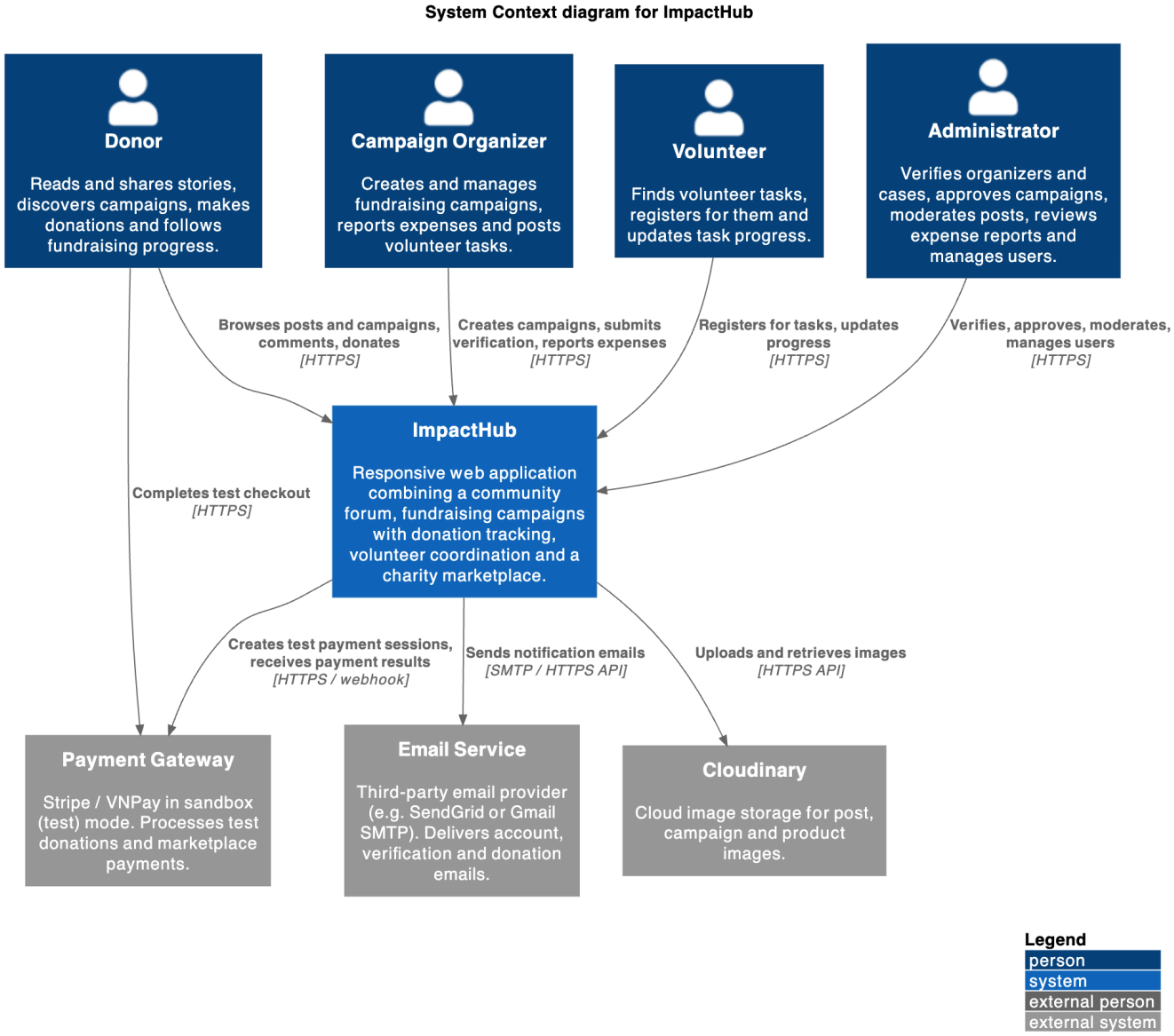
Out of scope:

- ✔ Real-money transactions, payouts or withdrawals to bank accounts
- ✔ Official identity verification, legal checks and background checks of organizers
- ✔ Quality inspection or authentication of marketplace products, and real shipping or delivery
- ✔ Real-time push updates such as WebSocket or live chat (progress updates after each donation and on page reload)
- ✔ Complex volunteer management such as scheduling, shift planning or location tracking
- ✔ Advanced financial management such as refunds, invoices, tax receipts or accounting reports

3 Proposed Solution

3.1 Preliminary Architecture

[Draw a C4 System Context diagram: the system, its users and external systems (payment gateway, email service, ...). Detailed architecture is presented in PA3.]



3.2 Technology Stack

Layer	Technology	Reason for choice
Front end	React (Vite) + Tailwind CSS	Component-based, fast to build responsive layouts, familiar to most members
Back end	Node.js + Express (REST API)	Same language (JavaScript) as the frontend, so all members can work across the stack
Database	PostgreSQL	Relational data (users, campaigns, donations) and transaction support to keep donation totals consistent
Hosting / deployment	Vercel (frontend), Render (backend + database)	Free tiers, easy deployment from GitHub
Testing & CI	GitHub (CI), Postman (API testing), Playwright (frontend test)	Supports automated workflows and efficient API validation.

3.3 Hardware and Operating Environment

1. Development Environment

Hardware: Standard developer laptops or PCs (Windows, macOS, or Linux) with a stable internet connection.

Software & Tools:

Code Editors / IDEs: Visual Studio Code.

UI/UX Design & Prototyping: Figma.

Version Control: Git & GitHub.

Testing & API Tools: Postman, modern web browsers (Google Chrome, Firefox, Edge).

2. Target Operating Environment

Client Devices: Any end-user device (Desktop, Laptop, Tablet, or Smartphone) equipped with a screen and standard input devices (mouse, keyboard, touch).

Supported Web Browsers: Modern versions of Google Chrome, Mozilla Firefox, Apple Safari, and Microsoft Edge (supporting HTML5, CSS3, and JavaScript ES6+).

Cloud Infrastructure & Hosting:

Frontend Deployment: Vercel cloud platform.

Backend & Database Hosting: Render cloud services (Node.js runtime environment and PostgreSQL database).

Third-party Services: Cloudinary (cloud storage for images) and Payment Gateway Sandbox environments (Stripe/VNPay).

4 Project Plan

4.1 Process and Phase Gates

The project follows the waterfall model. Each phase ends with a phase gate (a PA submission). After a PA is submitted it becomes a baseline; later changes must be updated in the document and listed in the next PA.

PA	Phase	Due	Exit criteria (fill in)
PA1	Project planning	End of week 3	The Project Proposal & Plan is completed. The project vision, target users, five Must-have features, and out-of-scope features are clearly defined. The proposed solution, technology stack, WBS, effort estimation, schedule, team responsibilities, cost estimate, risk plan, and collaboration tools are documented and reviewed by the team.
PA2	Requirements	End of week 5	The requirements document is completed for all five Must-have features. Functional and non-functional requirements, use cases, main and alternative flows, and acceptance criteria are defined. UI mockups are prepared, and the Requirements Traceability Matrix (RTM) connects requirements to use cases. The team reviews the requirements and resolves unclear points before moving forward.
	Prototype completion	End of week 6	A basic version of ImpactHub is available for testing. Main user flows, including login, forum posting, campaign creation, administrator approval, and simulated donations, can be demonstrated. The main parts of

			the system are connected, and basic tests confirm that these flows work. The interface and other details may still need improvement.
PA3	System Design	End of week 7	The system design documents are completed based on PA2 requirements. They include architecture diagrams, class and sequence diagrams, database design, and UI/UX designs. Key design decisions are recorded, and the RTM is updated to link requirements with the system design. The team reviews the designs before final implementation.
PA4	Test planning	End of week 8	The test plan is completed, including testing goals, scope, methods, tools, environment, and responsibilities. Test cases are prepared for all five Must-have features, covering normal use, invalid inputs, and access permissions. Each test case includes test data and expected results. The RTM is updated to ensure all Must-have requirements are covered by tests.
	Full Product Completion	End of week 10	All five Must-have features are implemented and connected. Users can manage accounts, interact with forum posts, create and manage campaigns, perform administrator approval actions, and make simulated donations. Donation totals and progress displays work correctly. The application is deployed, main workflows are tested, and critical bugs are fixed.
PA5	Implementation, testing & deployment	End of week 10	The completed application and source code are submitted with implementation evidence. Test cases are executed, bugs are

			recorded and reviewed, and automated tests cover at least two use cases with two scenarios each. The deployed website is checked, and test results, release notes, user guide, project review, and a demonstration video of no more than 10 minutes are completed.
Final	Presentation & demo	Week 11	The team completes the final presentation and demonstrates the deployed ImpactHub website. The demo shows the main features, especially the forum, campaign approval, and simulated donation process. Project results, limitations, and lessons learned are presented, and all final submission materials are checked.

4.2 Work Breakdown Structure (WBS)

Phase	Deliverable	Work Package
1 Initiation & Planning	1.1 Project Vision	1.1.1 Describe the preliminary problem and motivation of ImpactHub
		1.1.2 Identify donors, campaign organizers, volunteers and administrators
		1.1.3 Prepare the product position statement
		1.1.4 Define Must-have features and out-of-scope activities
	1.2 Solution	1.2.1 Draw the preliminary System Context Diagram (C4 level 1)
		1.2.2 Select the proposed frontend, backend and database technologies

		1.2.3 Describe hardware and operating environments
		1.2.4 Outline the hosting and deployment environment
	1.3 Project Plan	1.3.1 Define phase gates and exit criteria for PA1 to PA5 and Final
		1.3.2 Prepare the Work Breakdown Structure (WBS)
		1.3.3 Estimate person-hours for each work package
		1.3.4 Prepare the Gantt chart with dependencies and milestones
		1.3.5 Assign team roles and responsibilities
		1.3.6 Estimate human resources and project costs
	1.4 Risk Management	1.4.1 Identify technology and integration risks
		1.4.2 Identify requirements and scope risks
		1.4.3 Identify schedule and team risks
		1.4.4 Estimate risk exposure and document mitigation actions
	1.5 Tools & Collaboration Setup	1.5.1 Set up the communication channel and team meetings
		1.5.2 Set up the task board and assign initial work packages
		1.5.3 Set up the Git repository and review rules

		1.5.4 Create documentation folders and weekly report schedule
2 Requirements	2.1 Introduction & Problem Statement	2.1.1 Define document purpose, product scope and intended readers
		2.1.2 Prepare the project glossary
		2.1.3 Expand the problem statement and operating constraints
	2.2 Stakeholders	2.2.1 Describe donors and their interests
		2.2.2 Describe campaign organizers and their interests
		2.2.3 Describe administrators and their interests
		2.2.4 Describe volunteers as a non-Must-have user group
	2.3 Requirements	2.3.1 Write functional requirements for authentication and user roles
		2.3.2 Write functional requirements for community forum
		2.3.3 Write functional requirements for administrator approval actions
		2.3.4 Write functional requirements for fundraising campaigns
		2.3.5 Write functional requirements for simulated donations and progress tracking

		2.3.6 Specify measurable performance, security and usability requirements
		2.3.7 Document business rules for administrator permissions and campaign status changes
		2.3.8 Document rules for recording simulated donations without real-money transfers
		2.3.9 Write Given/When/Then acceptance criteria for Must-have features
		2.3.10 Assign MoSCoW priorities and requirement sources
	2.4 Use Case Model & Specifications	2.4.1 Identify actors and list use cases
		2.4.2 Draw the Use Case Diagram
		2.4.3 Specify registration and login use cases
		2.4.4 Specify forum post and interaction use cases
		2.4.5 Specify campaign creation and editing use cases
		2.4.6 Specify administrator approval use cases
		2.4.7 Specify simulated donation and progress tracking use cases
		2.4.8 Review alternative and exception flows for all Must-have use cases
	2.5 Prototype/Mockups	2.5.1 Prepare the screen list linked to use cases

		2.5.2 Create wireframes for login and profile screens
		2.5.3 Create wireframes for forum screens
		2.5.4 Create wireframes for campaign and donation screens
		2.5.5 Create wireframes for administrator review screens
		2.5.6 Link the interactive prototype and check basic accessibility
	2.6 Traceability & Requirements Review	2.6.1 Prepare the Requirements Traceability Matrix (RTM)
		2.6.2 Conduct the requirements review and record resolutions
		2.6.3 Confirm the requirements baseline
3 Design	3.1 Design Overview	3.1.1 Identify design goals and constraints based on high-priority requirements
	3.2 Architectural Design	3.2.1 Select and justify the architectural style
		3.2.2 Update the System Context Diagram (C4 level 1)
		3.2.3 Draw the Container Diagram (C4 level 2)
		3.2.4 Draw the Component Diagram for the main modules
		3.2.5 Draw the Deployment Diagram

		3.2.6 Record at least three Architecture Decision Records (ADRs)
	3.3 Object-Oriented Design	3.3.1 Draw the conceptual Domain Model
		3.3.2 Draw the design-level Class Diagram
		3.3.3 Specify six to eight important classes
		3.3.4 Draw Sequence Diagrams for three to five key use cases
		3.3.5 Draw State Machine Diagrams for campaign approval and donation status where needed
		3.3.6 Describe design principles and patterns used
	3.4 Data Design	3.4.1 Draw the Entity Relationship Diagram (ERD) for stored data
		3.4.2 Specify users, roles and profile data
		3.4.3 Specify forum posts, comments and reactions data
		3.4.4 Specify campaigns and approval status data
		3.4.5 Specify simulated donations and funding progress data
		3.4.6 Review normalisation and indexes for frequent queries
	3.5 User Interface and User Experience Design	3.5.1 Prepare the screen list and screen flow diagram

		3.5.2 Specify five to six important screens and their controls
		3.5.3 Prepare design images for the remaining screens
		3.5.4 Check contrast, keyboard access, form labels and responsive layouts
	3.6 Traceability Update & Requirements Changes	3.6.1 Extend the Requirements Traceability Matrix with components, classes and screens
		3.6.2 Record any changes to requirements since PA2
4 Test Planning	4.1 Test Plan	4.1.1 Define test objectives, scope and excluded features
		4.1.2 Select test levels and test types
		4.1.3 Describe test design techniques
		4.1.4 Specify the test environment and test data
		4.1.5 Define entry and exit criteria
		4.1.6 Document defect management and responsibilities
		4.1.7 Plan test schedule, tools and test risks
	4.2 Test Design	4.2.1 Identify equivalence partitions and boundary values for login and donations
		4.2.2 Prepare decision tables for user permissions and campaign status

		4.2.3 Review test design against Must-have acceptance criteria
	4.3 Test Cases	4.3.1 List and prioritise test cases for Must-have features
		4.3.2 Specify authentication and role-access test cases
		4.3.3 Specify forum post, comment and search test cases
		4.3.4 Specify administrator approval test cases
		4.3.5 Specify campaign creation and management test cases
		4.3.6 Specify simulated donation and progress test cases
		4.3.7 Review expected results and test data for each case
	4.4 Requirements – Test Traceability	4.4.1 Map each Must-have requirement to at least one test case
		4.4.2 Review coverage gaps and update the traceability matrix
5. Implementation, Testing & Deployment	5.1 Environment Setup	5.1.1 Set up the frontend and backend development environments
		5.1.2 Configure the PostgreSQL database connection
		5.1.3 Prepare test data and shared development conventions
		5.1.4 Configure Continuous Integration (CI) for code checks

	5.2 Frontend Development	5.2.1 Implement navigation and responsive page layouts
		5.2.2 Implement login, registration and profile pages
		5.2.3 Implement forum feed, post detail and interaction screens
		5.2.4 Implement campaign listing, details and creation screens
		5.2.5 Implement simulated donation and fundraising progress screens
		5.2.6 Implement administrator review and approval screens
		5.2.7 Connect page actions to backend Application Programming Interface (API) endpoints
	5.3 Database Development	5.3.1 Create database tables for users and roles
		5.3.2 Create database tables for forum posts and interactions
		5.3.3 Create database tables for campaigns and approval status
		5.3.4 Create database tables for simulated donations
		5.3.5 Prepare sample data and verify table relationships
	5.4 Backend – Authentication & User Roles	5.4.1 Implement registration and password login endpoints
		5.4.2 Implement JSON Web Token (JWT) authentication

		5.4.3 Implement role-based access checks
		5.4.4 Implement user profile viewing and editing
	5.5 Backend – Community Forum	5.5.1 Implement creating, editing and deleting forum posts
		5.5.2 Implement post image upload
		5.5.3 Implement comments and reactions
		5.5.4 Implement post search and sharing links
		5.5.5 Implement administrator post moderation actions
	5.6 Backend – Campaign Management	5.6.1 Implement campaign creation and editing endpoints
		5.6.2 Implement campaign listing and detail endpoints
		5.6.3 Implement campaign goal and status updates
		5.6.4 Restrict campaign editing to authorized users
	5.7 Backend – Administrator Approval	5.7.1 Implement administrator-only campaign approval and rejection
		5.7.2 Store campaign approval status and updates
		5.7.3 Restrict approval actions by user role
		5.7.4 Display approval results to organizers without claiming identity or legal verification

	5.8 Backend – Simulated Donations	5.8.1 Implement simulated donation submission
		5.8.2 Record donation results without real-money transfers
		5.8.3 Calculate and display fundraising totals and progress
		5.8.4 Prevent failed simulated donations from changing totals
	5.9 Integration & Feature Status	5.9.1 Integrate frontend, backend and database modules
		5.9.2 Review Must-have feature completion status
		5.9.3 Record deviations from the design
		5.9.4 Document source code structure and code review evidence
		5.9.5 Review implementation evidence and contribution records
	5.10 Testing & Continuous Integration	5.10.1 Run planned functional test cases and record results
		5.10.2 Write unit tests and record coverage
		5.10.3 Automate at least two use cases with two scenarios each
		5.10.4 Run automated checks through Continuous Integration (CI)
		5.10.5 Log defects and verify fixes
		5.10.6 Prepare the test execution summary and static analysis results

	5.11 Deployment & Release	5.11.1 Configure the deployment environment
		5.11.2 Deploy the frontend, backend and database
		5.11.3 Verify the working application in the deployment environment
		5.11.4 Prepare release notes and user guide
		5.11.5 Record a working software video of no more than ten minutes
	5.12 Project Retrospective & Presentation	5.12.1 Document achievements and remaining limitations
		5.12.2 Summarise lessons learned and improvement actions
		5.12.3 Prepare the PA5 project presentation
6 Final Presentation & Demo	6.1 Presentation Materials	6.1.1 Prepare final presentation slides
		6.1.2 Summarise scope, design, implementation and testing results
	6.2 Demo Preparation	6.2.1 Prepare demo accounts and sample campaigns
		6.2.2 Rehearse the forum, campaign approval and simulated donation flows
		6.2.3 Check deployed application and fallback demonstration material
	6.3 Final Review & Defense	6.3.1 Review final submission files and links

		6.3.2 Deliver the presentation and software demonstration

4.3 Estimation

A student-project scope is selected because ImpactHub is developed by a four-member academic team within an 11-week course. Estimates cover the five Must-have features, required documentation, testing, and a demonstrable deployment; optional features and production-grade services are not included. This keeps the planned workload achievable alongside coursework. Person-hours below are preliminary expert-judgement and analogy estimates for each Level-3 WBS work package.

WBS ID	Work package	Owner (role)	Effort (person-hours)	Method / assumptions
1.1.1	Describe the preliminary problem and motivation of ImpactHub	Business analyst	1	Expert judgement; focused review or planning task
1.1.2	Identify donors, campaign organizers, volunteers and administrators	Business analyst	1	Expert judgement; focused review or planning task
1.1.3	Prepare the product position statement	Business analyst	2	Expert judgement; concise course deliverable
1.1.4	Define Must-have features and out-of-scope activities	Business analyst	1	Expert judgement; focused review or planning task
1.2.1	Draw the preliminary System Context Diagram (C4 level 1)	Designer	3	Analogy; small academic design deliverable
1.2.2	Select the proposed frontend, backend and database technologies	Developer	1	Expert judgement; focused review or planning task
1.2.3	Describe hardware and operating environments	Developer	1	Expert judgement; focused review or planning task

1.2.4	Outline the hosting and deployment environment	Developer	4	Expert judgement; student implementation and integration effort
1.3.1	Define phase gates and exit criteria for PA1 to PA5 and Final	Project Manager	1	Expert judgement; focused review or planning task
1.3.2	Prepare the Work Breakdown Structure (WBS)	Project Manager	2	Expert judgement; concise course deliverable
1.3.3	Estimate person-hours for each work package	Project Manager	1	Expert judgement; focused review or planning task
1.3.4	Prepare the Gantt chart with dependencies and milestones	Project Manager	2	Expert judgement; concise course deliverable
1.3.5	Assign team roles and responsibilities	Project Manager	1	Expert judgement; focused review or planning task
1.3.6	Estimate human resources and project costs	Project Manager	1	Expert judgement; focused review or planning task
1.4.1	Identify technology and integration risks	Developer	1	Expert judgement; focused review or planning task
1.4.2	Identify requirements and scope risks	Business analyst	1	Expert judgement; focused review or planning task
1.4.3	Identify schedule and team risks	Project Manager	1	Expert judgement; focused review or planning task
1.4.4	Estimate risk exposure and document mitigation actions	Project Manager	2	Expert judgement; concise course deliverable
1.5.1	Set up the communication channel and team meetings	Project Manager	1	Expert judgement; focused review or planning task
1.5.2	Set up the task board and assign initial work packages	Project Manager	1	Expert judgement; focused review or planning task
1.5.3	Set up the Git repository and review rules	Developer	1	Expert judgement; focused review or planning task
1.5.4	Create documentation folders and weekly report schedule	Project Manager	2	Expert judgement; concise course deliverable
2.1.1	Define document purpose, product scope and intended readers	Business analyst	2	Expert judgement; concise course deliverable
2.1.2	Prepare the project glossary	Business analyst	2	Expert judgement; concise course deliverable

2.1.3	Expand the problem statement and operating constraints	Business analyst	1	Expert judgement; focused review or planning task
2.2.1	Describe donors and their interests	Business analyst	1	Expert judgement; focused review or planning task
2.2.2	Describe campaign organizers and their interests	Business analyst	1	Expert judgement; focused review or planning task
2.2.3	Describe administrators and their interests	Business analyst	1	Expert judgement; focused review or planning task
2.2.4	Describe volunteers as a non-Must-have user group	Business analyst	1	Expert judgement; focused review or planning task
2.3.1	Write functional requirements for authentication and user roles	Business analyst	2	Expert judgement; concise course deliverable
2.3.2	Write functional requirements for community forum	Business analyst	2	Expert judgement; concise course deliverable
2.3.3	Write functional requirements for administrator approval actions	Business analyst	2	Expert judgement; concise course deliverable
2.3.4	Write functional requirements for fundraising campaigns	Business analyst	2	Expert judgement; concise course deliverable
2.3.5	Write functional requirements for simulated donations and progress tracking	Business analyst	2	Expert judgement; concise course deliverable
2.3.6	Specify measurable performance, security and usability requirements	Business analyst	2	Expert judgement; concise course deliverable
2.3.7	Document business rules for administrator permissions and campaign status changes	Business analyst	2	Expert judgement; concise course deliverable
2.3.8	Document rules for recording simulated	Business analyst	2	Expert judgement; concise course deliverable

	donations without real-money transfers			
2.3.9	Write Given/When/Then acceptance criteria for Must-have features	Business analyst	2	Expert judgement; concise course deliverable
2.3.10	Assign MoSCoW priorities and requirement sources	Business analyst	1	Expert judgement; focused review or planning task
2.4.1	Identify actors and list use cases	Business analyst	1	Expert judgement; focused review or planning task
2.4.2	Draw the Use Case Diagram	Business analyst	3	Analogy; small academic design deliverable
2.4.3	Specify registration and login use cases	Business analyst	2	Expert judgement; concise course deliverable
2.4.4	Specify forum post and interaction use cases	Business analyst	2	Expert judgement; concise course deliverable
2.4.5	Specify campaign creation and editing use cases	Business analyst	2	Expert judgement; concise course deliverable
2.4.6	Specify administrator approval use cases	Business analyst	2	Expert judgement; concise course deliverable
2.4.7	Specify simulated donation and progress tracking use cases	Business analyst	2	Expert judgement; concise course deliverable
2.4.8	Review alternative and exception flows for all Must-have use cases	Business analyst	1	Expert judgement; focused review or planning task
2.5.1	Prepare the screen list linked to use cases	Designer	2	Expert judgement; concise course deliverable
2.5.2	Create wireframes for login and profile screens	Designer	3	Analogy; small academic design deliverable
2.5.3	Create wireframes for forum screens	Designer	3	Analogy; small academic design deliverable
2.5.4	Create wireframes for campaign and donation screens	Designer	3	Analogy; small academic design deliverable
2.5.5	Create wireframes for administrator review screens	Designer	3	Analogy; small academic design deliverable

2.5.6	Create basic interactive navigation and clickable Figma links	Designer	2	Expert judgement; concise course deliverable
2.6.1	Prepare the Requirements Traceability Matrix (RTM)	Business analyst	2	Expert judgement; concise course deliverable
2.6.2	Conduct the requirements review and record resolutions	Business analyst	1	Expert judgement; focused review or planning task
2.6.3	Confirm the requirements baseline	Business analyst	1	Expert judgement; focused review or planning task
3.1.1	Identify design goals and constraints based on high-priority requirements	Designer	1	Expert judgement; focused review or planning task
3.2.1	Select and justify the architectural style	Designer	1	Expert judgement; focused review or planning task
3.2.2	Update the System Context Diagram (C4 level 1)	Designer	3	Analogy; small academic design deliverable
3.2.3	Draw the Container Diagram (C4 level 2)	Designer	3	Analogy; small academic design deliverable
3.2.4	Draw the Component Diagram for the main modules	Designer	3	Analogy; small academic design deliverable
3.2.5	Draw the Deployment Diagram	Designer	4	Expert judgement; student implementation and integration effort
3.2.6	Record at least three Architecture Decision Records (ADRs)	Designer	1	Expert judgement; focused review or planning task
3.3.1	Draw the conceptual Domain Model	Designer	2	Expert judgement; concise course deliverable
3.3.2	Draw the design-level Class Diagram	Designer	3	Analogy; small academic design deliverable
3.3.3	Specify six to eight important classes	Designer	2	Expert judgement; concise course deliverable
3.3.4	Draw Sequence Diagrams for three to five key use cases	Designer	3	Analogy; small academic design deliverable
3.3.5	Draw State Machine Diagrams for campaign	Designer	3	Analogy; small academic design deliverable

	approval and donation status where needed			
3.3.6	Describe design principles and patterns used	Designer	1	Expert judgement; focused review or planning task
3.4.1	Draw the Entity Relationship Diagram (ERD) for stored data	Designer	3	Analogy; small academic design deliverable
3.4.2	Specify users, roles and profile data	Designer	2	Expert judgement; concise course deliverable
3.4.3	Specify forum posts, comments and reactions data	Designer	2	Expert judgement; concise course deliverable
3.4.4	Specify campaigns and approval status data	Designer	2	Expert judgement; concise course deliverable
3.4.5	Specify simulated donations and funding progress data	Designer	2	Expert judgement; concise course deliverable
3.4.6	Review normalisation and indexes for frequent queries	Designer	1	Expert judgement; focused review or planning task
3.5.1	Prepare the screen list and screen flow diagram	Designer	3	Analogy; small academic design deliverable
3.5.2	Specify five to six important screens and their controls	Designer	2	Expert judgement; concise course deliverable
3.5.3	Create high-fidelity designs for key screens	Designer	2	Expert judgement; concise course deliverable
3.5.4	Integrate and refine interactive prototype flows	Designer	4	Expert judgement; student implementation and integration effort
3.6.1	Extend the Requirements Traceability Matrix with components, classes and screens	Business analyst	1	Expert judgement; focused review or planning task
3.6.2	Record any changes to requirements since PA2	Business analyst	1	Expert judgement; focused review or planning task
4.1.1	Define test objectives, scope and excluded features	Tester	1	Expert judgement; focused review or planning task
4.1.2	Select test levels and test types	Tester	1	Expert judgement; focused review or planning task
4.1.3	Describe test design techniques	Tester	1	Expert judgement; focused review or planning task

4.1.4	Specify the test environment and test data	Tester	2	Expert judgement; concise course deliverable
4.1.5	Define entry and exit criteria	Business analyst	1	Expert judgement; focused review or planning task
4.1.6	Document defect management and responsibilities	Business analyst	2	Expert judgement; concise course deliverable
4.1.7	Plan test schedule, tools and test risks	Tester	1	Expert judgement; focused review or planning task
4.2.1	Identify equivalence partitions and boundary values for login and donations	Business analyst	1	Expert judgement; focused review or planning task
4.2.2	Prepare decision tables for user permissions and campaign status	Business analyst	2	Expert judgement; concise course deliverable
4.2.3	Review test design against Must-have acceptance criteria	Tester	1	Expert judgement; focused review or planning task
4.3.1	List and prioritise test cases for Must-have features	Tester	3	Analogy; limited Must-have test coverage
4.3.2	Specify authentication and role-access test cases	Tester	3	Analogy; limited Must-have test coverage
4.3.3	Specify forum post, comment and search test cases	Tester	3	Analogy; limited Must-have test coverage
4.3.4	Specify administrator approval test cases	Tester	3	Analogy; limited Must-have test coverage
4.3.5	Specify campaign creation and management test cases	Tester	3	Analogy; limited Must-have test coverage
4.3.6	Specify simulated donation and progress test cases	Tester	3	Analogy; limited Must-have test coverage
4.3.7	Review expected results and test data for each case	Tester	1	Expert judgement; focused review or planning task
4.4.1	Map each Must-have requirement to at least one test case	Tester	1	Expert judgement; focused review or planning task

4.4.2	Review coverage gaps and update the traceability matrix	Business analyst	1	Expert judgement; focused review or planning task
5.1.1	Set up the frontend and backend development environments	Developer	4	Expert judgement; student implementation and integration effort
5.1.2	Configure the PostgreSQL database connection	Developer	4	Expert judgement; student implementation and integration effort
5.1.3	Prepare test data and shared development conventions	Tester	4	Expert judgement; student implementation and integration effort
5.1.4	Configure Continuous Integration (CI) for code checks	Developer	4	Expert judgement; student implementation and integration effort
5.2.1	Implement navigation and responsive page layouts	Developer	4	Expert judgement; student implementation and integration effort
5.2.2	Implement login, registration and profile pages	Developer	4	Expert judgement; student implementation and integration effort
5.2.3	Implement forum feed, post detail and interaction screens	Developer	4	Expert judgement; student implementation and integration effort
5.2.4	Implement campaign listing, details and creation screens	Developer	4	Expert judgement; student implementation and integration effort
5.2.5	Implement simulated donation and fundraising progress screens	Developer	4	Expert judgement; student implementation and integration effort
5.2.6	Implement administrator review and approval screens	Developer	4	Expert judgement; student implementation and integration effort
5.2.7	Connect page actions to backend Application Programming Interface (API) endpoints	Developer	1	Expert judgement; focused review or planning task

5.3.1	Create database tables for users and roles	Developer	4	Expert judgement; student implementation and integration effort
5.3.2	Create database tables for forum posts and interactions	Developer	4	Expert judgement; student implementation and integration effort
5.3.3	Create database tables for campaigns and approval status	Developer	4	Expert judgement; student implementation and integration effort
5.3.4	Create database tables for simulated donations	Developer	4	Expert judgement; student implementation and integration effort
5.3.5	Prepare sample data and verify table relationships	Developer	2	Expert judgement; concise course deliverable
5.4.1	Implement registration and password login endpoints	Developer	4	Expert judgement; student implementation and integration effort
5.4.2	Implement JSON Web Token (JWT) authentication	Developer	4	Expert judgement; student implementation and integration effort
5.4.3	Implement role-based access checks	Developer	4	Expert judgement; student implementation and integration effort
5.4.4	Implement user profile viewing and editing	Developer	4	Expert judgement; student implementation and integration effort
5.5.1	Implement creating, editing and deleting forum posts	Developer	4	Expert judgement; student implementation and integration effort
5.5.2	Implement post image upload	Developer	4	Expert judgement; student implementation and integration effort
5.5.3	Implement comments and reactions	Developer	4	Expert judgement; student implementation and integration effort
5.5.4	Implement post search and sharing links	Developer	4	Expert judgement; student implementation and integration effort

5.5.5	Implement administrator post moderation actions	Developer	4	Expert judgement; student implementation and integration effort
5.6.1	Implement campaign creation and editing endpoints	Developer	4	Expert judgement; student implementation and integration effort
5.6.2	Implement campaign listing and detail endpoints	Developer	4	Expert judgement; student implementation and integration effort
5.6.3	Implement campaign goal and status updates	Developer	4	Expert judgement; student implementation and integration effort
5.6.4	Restrict campaign editing to authorized users	Developer	1	Expert judgement; focused review or planning task
5.7.1	Implement administrator-only campaign approval and rejection	Developer	4	Expert judgement; student implementation and integration effort
5.7.2	Store campaign approval status and updates	Developer	1	Expert judgement; focused review or planning task
5.7.3	Restrict approval actions by user role	Developer	1	Expert judgement; focused review or planning task
5.7.4	Display approval results to organizers without claiming identity or legal verification	Developer	1	Expert judgement; focused review or planning task
5.8.1	Implement simulated donation submission	Developer	4	Expert judgement; student implementation and integration effort
5.8.2	Record donation results without real-money transfers	Developer	1	Expert judgement; focused review or planning task
5.8.3	Calculate and display fundraising totals and progress	Developer	1	Expert judgement; focused review or planning task
5.8.4	Prevent failed simulated donations from changing totals	Developer	1	Expert judgement; focused review or planning task

5.9.1	Integrate frontend, backend and database modules	Developer	4	Expert judgement; student implementation and integration effort
5.9.2	Review Must-have feature completion status	Business analyst	1	Expert judgement; focused review or planning task
5.9.3	Record deviations from the design	Business analyst	1	Expert judgement; focused review or planning task
5.9.4	Document source code structure and code review evidence	Business analyst	2	Expert judgement; concise course deliverable
5.9.5	Review implementation evidence and contribution records	Business analyst	4	Expert judgement; student implementation and integration effort
5.10.1	Run planned functional test cases and record results	Tester	3	Analogy; limited Must-have test coverage
5.10.2	Write unit tests and record coverage	Tester	3	Analogy; limited Must-have test coverage
5.10.3	Automate at least two use cases with two scenarios each	Business analyst	4	Expert judgement; student implementation and integration effort
5.10.4	Run automated checks through Continuous Integration (CI)	Business analyst	4	Expert judgement; student implementation and integration effort
5.10.5	Log defects and verify fixes	Business analyst	1	Expert judgement; focused review or planning task
5.10.6	Prepare the test execution summary and static analysis results	Tester	3	Analogy; limited Must-have test coverage
5.11.1	Configure the deployment environment	Developer	4	Expert judgement; student implementation and integration effort
5.11.2	Deploy the frontend, backend and database	Developer	4	Expert judgement; student implementation and integration effort
5.11.3	Verify the working application in the deployment environment	Business analyst	4	Expert judgement; student implementation and integration effort
5.11.4	Prepare release notes and user guide	Developer	2	Expert judgement; concise course deliverable

5.11.5	Record a working software video of no more than ten minutes	Business analyst	1	Expert judgement; focused review or planning task
5.12.1	Document achievements and remaining limitations	Business analyst	2	Expert judgement; concise course deliverable
5.12.2	Summarise lessons learned and improvement actions	Business analyst	1	Expert judgement; focused review or planning task
5.12.3	Prepare the PA5 project presentation	Business analyst	2	Expert judgement; concise course deliverable
6.1.1	Prepare final presentation slides	Project Manager	2	Expert judgement; concise course deliverable
6.1.2	Summarise scope, design, implementation and testing results	Tester	4	Expert judgement; student implementation and integration effort
6.2.1	Prepare demo accounts and sample campaigns	Developer	2	Expert judgement; concise course deliverable
6.2.2	Rehearse the forum, campaign approval and simulated donation flows	Project Manager	1	Expert judgement; focused review or planning task
6.2.3	Check deployed application and fallback demonstration material	Developer	4	Expert judgement; student implementation and integration effort
6.3.1	Review final submission files and links	Business analyst	1	Expert judgement; focused review or planning task
6.3.2	Deliver the presentation and software demonstration	Project Manager	1	Expert judgement; focused review or planning task

4.4 Schedule

Detailed Gantt chart schedule, PA timeline, milestones (Prototype, Final), critical path are stated with the images down below according to the instructor's guide and template provided. For a better resolution / visualization, please check the attached file in the submission or visit the link.

Work Breakdown Structure With Gantt Chart Template

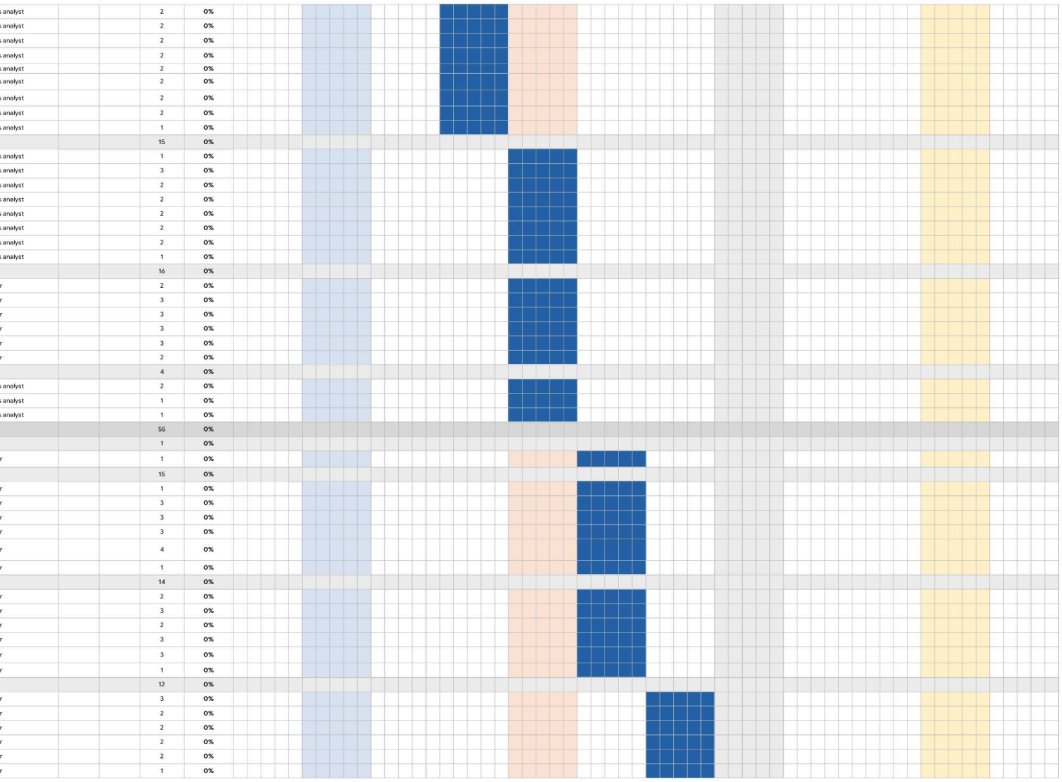
Project Title	ImpactHub
Project Manager	Trần Anh Khoa
Company Name	Matcha kem dâu
Date	10/16/2026

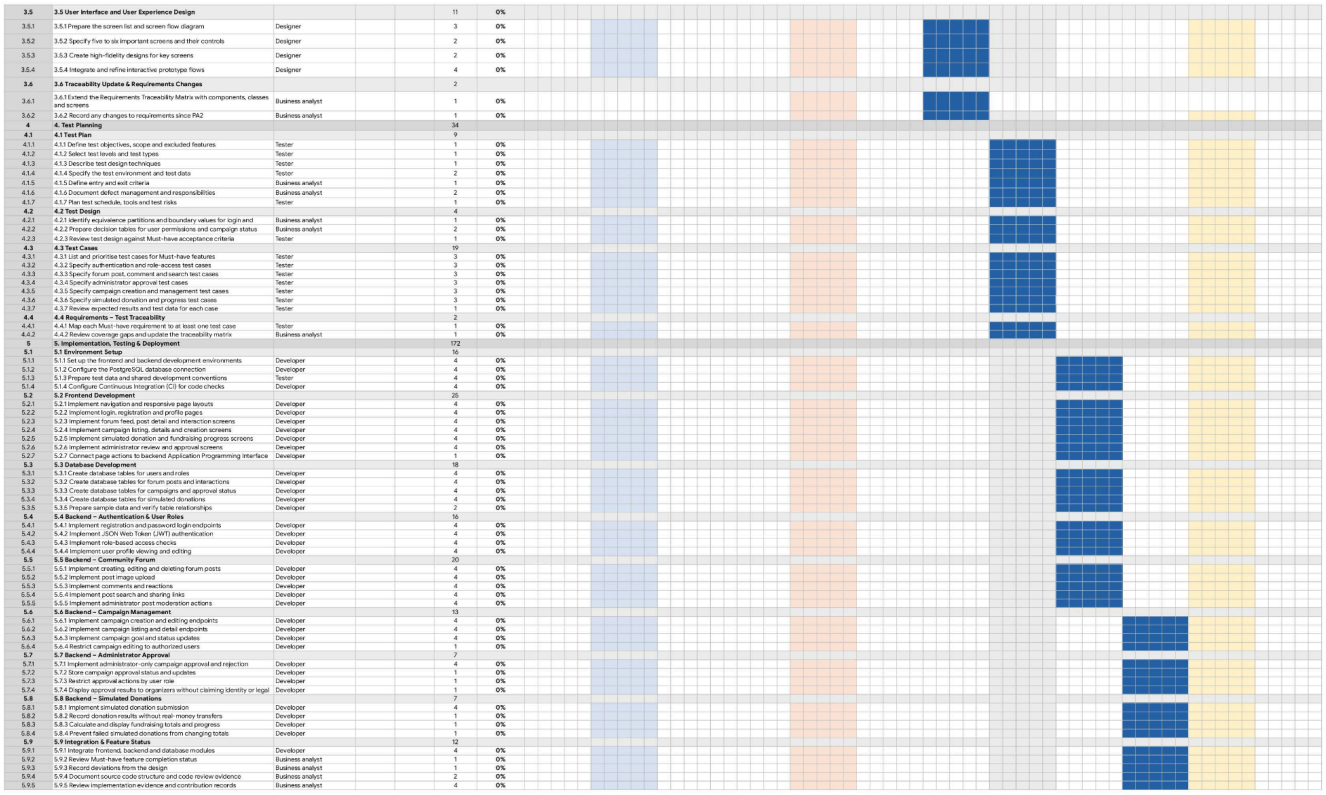
*Note: Darker blue (#2563a6) indicate the critical path. Important milestone is colored red in the PA timeline

PA1 | PA2 | PA3 - Prototype - W6-W7 | PA4 - 30/11-04/12 W8 | PA5 - 07/12-18/12 - W9-W10 | Final + Defense - W11-W12

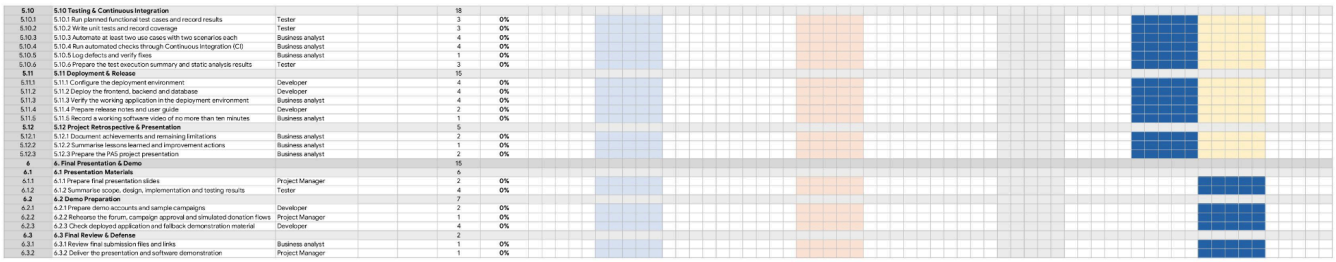
WBS Number	Task Title	Task Owner	Start Date	Due Date	Duration	% of Task Complete	Week 1	Week 2	Week 3 (12-18/10)	Week 4 (19-25/10)	Week 5 (26/10-1/11)	Week 6 (2-8/11)	Week 7 (9-15/11)	Week 8 (16-22/11)	Week 9 (23-29/11)	Week 10 (30/11-6/12)	Week 11 (7-13/12)	Week 12 (14-20/12)
							M T W R F	M T W R F	M T W R F	M T W R F	M T W R F	M T W R F	M T W R F	M T W R F	M T W R F	M T W R F	M T W R F	M T W R F
1	**1. Initiation & Planning**				32													
1.1	**1.1 Project Vision**				5	100%												
1.1.1	1.1.1 Describe the preliminary problem and motivation of ImpactHub	Business analyst			1	100%												
1.1.2	1.1.2 Identify donors, campaign organizers, volunteers and administrators	Business analyst			1	100%												
1.1.3	1.1.3 Prepare the product position statement	Business analyst			2	100%												
1.1.4	1.1.4 Define Must-have features and out-of-scope activities	Business analyst			1	100%												
1.2	**1.2 Solution**				9	100%												
1.2.1	1.2.1 Draw the preliminary System Context Diagram (C4 level 1)	Designer			3	100%												
1.2.2	1.2.2 Select the proposed frontend, backend and database technologies	Developer			1	100%												
1.2.3	1.2.3 Describe hardware and operating environments	Developer			1	100%												
1.2.4	1.2.4 Outline the hosting and deployment environment	Developer			4	100%												
1.3	**1.3 Project Plan**				8	100%												
1.3.1	1.3.1 Define phase gates and exit criteria for PA1 to PA5 and Final	Project Manager			1	100%												
1.3.2	1.3.2 Prepare the Work Breakdown Structure (WBS)	Project Manager			2	100%												
1.3.3	1.3.3 Estimate person-hours for each work package	Project Manager			1	100%												
1.3.4	1.3.4 Prepare the Gantt chart with dependencies and milestones	Project Manager			2	100%												
1.3.5	1.3.5 Assign team roles and responsibilities	Project Manager			1	100%												
1.3.6	1.3.6 Estimate human resources and project costs	Project Manager			1	100%												
1.4	**1.4 Risk Management**				5	100%												
1.4.1	1.4.1 Identify technology and integration risks	Developer			1	100%												
1.4.2	1.4.2 Identify requirements and scope risks	Business analyst			1	100%												
1.4.3	1.4.3 Identify schedule and team risks	Project Manager			1	100%												
1.4.4	1.4.4 Estimate risk exposure and document mitigation actions	Project Manager			2	100%												
1.5	**1.5 Tools & Collaboration Setup**				5	0%												
1.5.1	1.5.1 Set up the communication channel and team meetings	Project Manager			1	100%												
1.5.2	1.5.2 Set up the task board and assign initial work packages	Project Manager			1	100%												
1.5.3	1.5.3 Set up the Git repository and review rules	Developer			1	100%												
1.5.4	1.5.4 Create documentation folders and weekly report schedule	Project Manager			2	100%												
2	**2. Requirements**				63	0%												
2.1	**2.1 Introduction & Problem Statement**				5	0%												
2.1.1	2.1.1 Define document purpose, product scope and intended readers	Business analyst			2	0%												
2.1.2	2.1.2 Prepare the project glossary	Business analyst			2	0%												
2.1.3	2.1.3 Expand the problem statement and operating constraints	Business analyst			1	0%												
2.2	**2.2 Stakeholders**				4	0%												
2.2.1	2.2.1 Describe donors and their interests	Business analyst			1	0%												
2.2.2	2.2.2 Describe campaign organizers and their interests	Business analyst			1	0%												
2.2.3	2.2.3 Describe administrators and their interests	Business analyst			1	0%												
2.2.4	2.2.4 Describe volunteers as a non-Must-have user group	Business analyst			1													
2.3	**2.3 Requirements**				19	0%												
2.3.1	2.3.1 Write functional requirements for authentication and user roles	Business analyst			2	0%												

2.3.2	2.3.2 Write functional requirements for community forum	Business analyst			2	0%
2.3.3	2.3.3 Write functional requirements for administrator approval actions	Business analyst			2	0%
2.3.4	2.3.4 Write functional requirements for fundraising campaigns	Business analyst			2	0%
2.3.5	2.3.5 Write functional requirements for simulated donations and progress tracking	Business analyst			2	0%
2.3.6	2.3.6 Specify measurable performance, security and usability requirements	Business analyst			2	0%
2.3.7	2.3.7 Document business rules for administrator permissions and campaign status changes	Business analyst			2	0%
2.3.8	2.3.8 Document rules for recording simulated donations without real-money transfers	Business analyst			2	0%
2.3.9	2.3.9 Write Given/When/Then acceptance criteria for Must-have features	Business analyst			2	0%
2.3.10	2.3.10 Assign MoSCoW priorities and requirement sources	Business analyst			1	0%
2.4	**2.4 Use Case Model & Specifications**				15	0%
2.4.1	2.4.1 Identify actors and list use cases	Business analyst			1	0%
2.4.2	2.4.2 Draw the Use Case Diagram	Business analyst			3	0%
2.4.3	2.4.3 Specify registration and login use cases	Business analyst			2	0%
2.4.4	2.4.4 Specify forum post and interaction use cases	Business analyst			2	0%
2.4.5	2.4.5 Specify campaign creation and editing use cases	Business analyst			2	0%
2.4.6	2.4.6 Specify administrator approval use cases	Business analyst			2	0%
2.4.7	2.4.7 Specify simulated donation and progress tracking use cases	Business analyst			2	0%
2.4.8	2.4.8 Review alternative and exception flows for all Must-have use cases	Business analyst			1	0%
2.5	**2.5 Prototype/Mockups**				16	0%
2.5.1	2.5.1 Prepare the screen list linked to use cases	Designer			2	0%
2.5.2	2.5.2 Create wireframes for login and profile screens	Designer			3	0%
2.5.3	2.5.3 Create wireframes for forum screens	Designer			3	0%
2.5.4	2.5.4 Create wireframes for campaign and donation screens	Designer			3	0%
2.5.5	2.5.5 Create wireframes for administrator review screens	Designer			3	0%
2.5.6	2.5.6 Create basic interactive navigation and clickable Figma links	Designer			2	0%
2.6	**2.6 Traceability & Requirements Review**				4	0%
2.6.1	2.6.1 Prepare the Requirements Traceability Matrix (RTM)	Business analyst			2	0%
2.6.2	2.6.2 Conduct the requirements review and record resolutions	Business analyst			1	0%
2.6.3	2.6.3 Confirm the requirements baseline	Business analyst			1	0%
3	**3. Design**				55	0%
3.1	**3.1 Design Overview**				1	0%
3.1.1	3.1.1 Identify design goals and constraints based on high-priority requirements	Designer			1	0%
3.2	**3.2 Architectural Design**				15	0%
3.2.1	3.2.1 Select and justify the architectural style	Designer			1	0%
3.2.2	3.2.2 Update the System Context Diagram (C4 level 1)	Designer			3	0%
3.2.3	3.2.3 Draw the Container Diagram (C4 level 2)	Designer			3	0%
3.2.4	3.2.4 Draw the Component Diagram for the main modules	Designer			3	0%
3.2.5	3.2.5 Draw the Deployment Diagram	Designer			4	0%
3.2.6	3.2.6 Record at least three Architecture Decision Records (ADRs)	Designer			1	0%
3.3	**3.3 Object-Oriented Design**				14	0%
3.3.1	3.3.1 Draw the conceptual Domain Model	Designer			2	0%
3.3.2	3.3.2 Draw the design-level Class Diagram	Designer			3	0%
3.3.3	3.3.3 Specify six to eight important classes	Designer			2	0%
3.3.4	3.3.4 Draw Sequence Diagrams for three to five key use cases	Designer			3	0%
3.3.5	3.3.5 Draw State Machine Diagrams for campaign approval and donation status where needed	Designer			3	0%
3.3.6	3.3.6 Describe design principles and patterns used	Designer			1	0%
3.4	**3.4 Data Design**				12	0%
3.4.1	3.4.1 Draw the Entity Relationship Diagram (ERD) for stored data	Designer			3	0%
3.4.2	3.4.2 Specify users, roles and profile data	Designer			2	0%
3.4.3	3.4.3 Specify forum posts, comments and reactions data	Designer			2	0%
3.4.4	3.4.4 Specify campaigns and approval status data	Designer			2	0%
3.4.5	3.4.5 Specify simulated donations and funding progress data	Designer			2	0%
3.4.6	3.4.6 Review normalisation and indexes for frequent queries	Designer			1	0%





3.5	**3.5 User Interface and User Experience Design**		11	0%
3.5.1	3.5.1 Prepare the screen list and screen flow diagram	Designer	3	0%
3.5.2	3.5.2 Specify five to six important screens and their controls	Designer	2	0%
3.5.3	3.5.3 Create high-fidelity designs for key screens	Designer	2	0%
3.5.4	3.5.4 Integrate and refine interactive prototype flows	Designer	4	0%
3.6	**3.6 Traceability Update & Requirements Changes**		2	
3.6.1	3.6.1 Extend the Requirements Traceability Matrix with components, classes and screens	Business analyst	1	0%
3.6.2	3.6.2 Record any changes to requirements since PA2	Business analyst	1	0%
4	**4. Test Planning**		34	
4.1	**4.1 Test Plan**		9	
4.1.1	4.1.1 Define test objectives, scope and excluded features	Tester	1	0%
4.1.2	4.1.2 Select test levels and test types	Tester	1	0%
4.1.3	4.1.3 Describe test design techniques	Tester	1	0%
4.1.4	4.1.4 Specify the test environment and test data	Tester	2	0%
4.1.5	4.1.5 Define entry and exit criteria	Business analyst	1	0%
4.1.6	4.1.6 Document defect management and responsibilities	Business analyst	2	0%
4.1.7	4.1.7 Plan test schedule, tools and test risks	Tester	1	0%
4.2	**4.2 Test Design**		4	
4.2.1	4.2.1 Identify equivalence partitions and boundary values for login and	Business analyst	1	0%
4.2.2	4.2.2 Prepare decision tables for user permissions and campaign status	Business analyst	2	0%
4.2.3	4.2.3 Review test design against Must-have acceptance criteria	Tester	1	0%
4.3	**4.3 Test Cases**		19	
4.3.1	4.3.1 List and prioritise test cases for Must-have features	Tester	3	0%
4.3.2	4.3.2 Specify authentication and role-access test cases	Tester	3	0%
4.3.3	4.3.3 Specify forum post, comment and search test cases	Tester	3	0%
4.3.4	4.3.4 Specify administrator approval test cases	Tester	3	0%
4.3.5	4.3.5 Specify campaign creation and management test cases	Tester	3	0%
4.3.6	4.3.6 Specify simulated donation and progress test cases	Tester	3	0%
4.3.7	4.3.7 Review expected results and test data for each case	Tester	1	0%
4.4	**4.4 Requirements – Test Traceability**		2	
4.4.1	4.4.1 Map each Must-have requirement to at least one test case	Tester	1	0%
4.4.2	4.4.2 Review coverage gaps and update the traceability matrix	Business analyst	1	0%
5	**5. Implementation, Testing & Deployment**		172	
5.1	**5.1 Environment Setup**		16	
5.1.1	5.1.1 Set up the frontend and backend development environments	Developer	4	0%
5.1.2	5.1.2 Configure the PostgreSQL database connection	Developer	4	0%
5.1.3	5.1.3 Prepare test data and shared development conventions	Tester	4	0%
5.1.4	5.1.4 Configure Continuous Integration (CI) for code checks	Developer	4	0%
5.2	**5.2 Frontend Development**		25	
5.2.1	5.2.1 Implement navigation and responsive page layouts	Developer	4	0%
5.2.2	5.2.2 Implement login, registration and profile pages	Developer	4	0%
5.2.3	5.2.3 Implement forum feed, post detail and interaction screens	Developer	4	0%
5.2.4	5.2.4 Implement campaign listing, details and creation screens	Developer	4	0%
5.2.5	5.2.5 Implement simulated donation and fundraising progress screens	Developer	4	0%
5.2.6	5.2.6 Implement administrator review and approval screens	Developer	4	0%
5.2.7	5.2.7 Connect page actions to backend Application Programming Interface	Developer	1	0%
5.3	**5.3 Database Development**		18	
5.3.1	5.3.1 Create database tables for users and roles	Developer	4	0%
5.3.2	5.3.2 Create database tables for forum posts and interactions	Developer	4	0%
5.3.3	5.3.3 Create database tables for campaigns and approval status	Developer	4	0%
5.3.4	5.3.4 Create database tables for simulated donations	Developer	4	0%
5.3.5	5.3.5 Prepare sample data and verify table relationships	Developer	2	0%
5.4	**5.4 Backend – Authentication & User Roles**		16	
5.4.1	5.4.1 Implement registration and password login endpoints	Developer	4	0%
5.4.2	5.4.2 Implement JSON Web Token (JWT) authentication	Developer	4	0%
5.4.3	5.4.3 Implement role-based access checks	Developer	4	0%
5.4.4	5.4.4 Implement user profile viewing and editing	Developer	4	0%
5.5	**5.5 Backend – Community Forum**		20	
5.5.1	5.5.1 Implement creating, editing and deleting forum posts	Developer	4	0%
5.5.2	5.5.2 Implement post image upload	Developer	4	0%
5.5.3	5.5.3 Implement comments and reactions	Developer	4	0%
5.5.4	5.5.4 Implement post search and sharing links	Developer	4	0%
5.5.5	5.5.5 Implement administrator post moderation actions	Developer	4	0%
5.6	**5.6 Backend – Campaign Management**		13	
5.6.1	5.6.1 Implement campaign creation and editing endpoints	Developer	4	0%
5.6.2	5.6.2 Implement campaign listing and detail endpoints	Developer	4	0%
5.6.3	5.6.3 Implement campaign goal and status updates	Developer	4	0%
5.6.4	5.6.4 Restrict campaign editing to authorized users	Developer	1	0%
5.7	**5.7 Backend – Administrator Approval**		7	
5.7.1	5.7.1 Implement administrator-only campaign approval and rejection	Developer	4	0%
5.7.2	5.7.2 Store campaign approval status and updates	Developer	1	0%
5.7.3	5.7.3 Restrict approval actions by user role	Developer	1	0%
5.7.4	5.7.4 Display approval results to organizers without claiming identity or legal	Developer	1	0%
5.8	**5.8 Backend – Simulated Donations**		7	
5.8.1	5.8.1 Implement simulated donation submission	Developer	4	0%
5.8.2	5.8.2 Record donation results without real-money transfers	Developer	1	0%
5.8.3	5.8.3 Calculate and display fundraising totals and progress	Developer	1	0%
5.8.4	5.8.4 Prevent failed simulated donations from changing totals	Developer	1	0%
5.9	**5.9 Integration & Feature Status**		12	
5.9.1	5.9.1 Integrate frontend, backend and database modules	Developer	4	0%
5.9.2	5.9.2 Review Must-have feature completion status	Business analyst	1	0%
5.9.3	5.9.3 Record deviations from the design	Business analyst	1	0%
5.9.4	5.9.4 Document source code structure and code review evidence	Business analyst	2	0%
5.9.5	5.9.5 Review implementation evidence and contribution records	Business analyst	4	0%



5.10	**5.10 Testing & Continuous Integration**		18	
5.10.1	5.10.1 Run planned functional test cases and record results	Tester	3	0%
5.10.2	5.10.2 Write unit tests and record coverage	Tester	3	0%
5.10.3	5.10.3 Automate at least two use cases with two scenarios each	Business analyst	4	0%
5.10.4	5.10.4 Run automated checks through Continuous Integration (CI)	Business analyst	4	0%
5.10.5	5.10.5 Log defects and verify fixes	Business analyst	1	0%
5.10.6	5.10.6 Prepare the test execution summary and static analysis results	Tester	3	0%
5.11	**5.11 Deployment & Release**		15	
5.11.1	5.11.1 Configure the deployment environment	Developer	4	0%
5.11.2	5.11.2 Deploy the frontend, backend and database	Developer	4	0%
5.11.3	5.11.3 Verify the working application in the deployment environment	Business analyst	4	0%
5.11.4	5.11.4 Prepare release notes and user guide	Developer	2	0%
5.11.5	5.11.5 Record a working software video of no more than ten minutes	Business analyst	1	0%
5.12	**5.12 Project Retrospective & Presentation**		5	
5.12.1	5.12.1 Document achievements and remaining limitations	Business analyst	2	0%
5.12.2	5.12.2 Summarise lessons learned and improvement actions	Business analyst	1	0%
5.12.3	5.12.3 Prepare the PA5 project presentation	Business analyst	2	0%
6	**6. Final Presentation & Demo**		15	
6.1	**6.1 Presentation Materials**		6	
6.1.1	6.1.1 Prepare final presentation slides	Project Manager	2	0%
6.1.2	6.1.2 Summarise scope, design, implementation and testing results	Tester	4	0%
6.2	**6.2 Demo Preparation**		7	
6.2.1	6.2.1 Prepare demo accounts and sample campaigns	Developer	2	0%
6.2.2	6.2.2 Rehearse the forum, campaign approval and simulated donation flows	Project Manager	1	0%
6.2.3	6.2.3 Check deployed application and fallback demonstration material	Developer	4	0%
6.3	**6.3 Final Review & Defense**		2	
6.3.1	6.3.1 Review final submission files and links	Business analyst	1	0%
6.3.2	6.3.2 Deliver the presentation and software demonstration	Project Manager	1	0%

4.5 Roles and Responsibilities

Role	Main	Backup	Responsibilities
Team lead / Project manager	Trần Anh Khoa	Nguyễn Trần Phương Thúy	Plan project activities, assign tasks, track progress, manage risks, prepare weekly reports and coordinate team members.
Business analyst	Nguyễn Trần Phương Thúy	Hồng Gia Bảo	Gather and document requirements, write user stories and use cases, define acceptance criteria and review requirements.

Designer / Architect	Hồng Gia Bảo	Lê Nho Duy Phước	Design UI/UX, prepare wireframes, support system architecture and review design consistency.
Developers	Lê Nho Duy Phước	Trần Anh Khoa	Develop frontend and backend features, manage the database, integrate modules, write unit tests and review code.
Tester	Trần Anh Khoa		Prepare test plans and test cases, execute tests, report bugs, verify fixes and summarize test results.

4.6 Human Resources and Cost Estimate

4.6.1 Human Resources

- The ImpactHub project has four team members. Each member is responsible for different tasks based on their assigned roles.
- The estimated working hours are calculated from the work packages in Section 4.3.

Main Role	Estimated Effort	Share of Total Effort
PM / Tester	63	17.17%
BA	90	24.52%
Designer	72	19.62%
Developer	142	38.69%
Total hour	367	100%

- The total estimated effort is 367 person-hours over 11 weeks.
- The workload is divided based on each member's responsibilities. Development tasks require more time, while documentation and design tasks have lower estimated effort.
- Team members may support each other when necessary to complete the project on schedule.

4.6.2 Human Resource Cost

- For this academic project, an assumed hourly rate of 80,000 VND is used for all team members.
- This rate is used to estimate the value of student work, not actual salaries.
- Human Resource Cost = Estimated Effort × Hourly Rate

Team Member	Estimated Hours	Hourly Rate (VND)	Estimated Cost
Trần Anh Khoa	63	80.000	5.040.000
Nguyễn Trần Phương Thúy	90	80.000	7.200.000
Hồng Gia Bảo	72	80.000	5.760.000
Lê Nho Duy Phước	142	80.000	11.360.000
Total	367		29.360.000

- The total estimated project cost is 29,360,000 VND, based on 367 person-hours of student work.

4.6.3 Tool Cost

Tool	Purpose	Plan	Estimated Cost (VND)
Visual Studio Code	Code editing	Free	0
Git & GitHub	Version control and CI	Free	0
Figma	UI/UX design	Free	0
Postman	API testing	Free	0
Playwright	Automated testing	Free / open-source	0
React, Vite, Tailwind CSS	Frontend development	Free / open-source	0
Node.js, Express, PostgreSQL	Backend and database software	Free / open-source	0
Total			0

4.6.4 Hosting Cost

Service	Purpose	Plan	Estimated Cost (VND)
Vercel	Frontend deployment	Free	0
Render	Backend hosting	Free allowance	0
Render PostgreSQL	Database hosting	Free allowance, if available	0
Cloudinary	Image storage	Free	0
Stripe / VNPay sandbox	Simulated payment testing	Test / sandbox	0
Total			0

4.6.5 Total Project Cost Summary

Cost Category	Estimated Cost (VND)
Human resources (estimated value of student work)	29,360,000
Tools	0
Hosting & cloud services	0
Total estimated project value	29,360,000
Actual planned cash expenditure for tools & hosting	0

- Note: All listed tools and hosting services are budgeted at 0 VND under free plans or free usage allowances. Paid upgrades, usage over free limits, and optional services are excluded from this estimate.

5 Risk Management

[Identify at least 5 project risks (technology, people, organisational, requirements, estimation). Risk exposure RE = Probability × Loss (use the scales from the Project Management lecture).]

ID	Risk	Type	Probability	Loss	RE	Mitigation / contingency	Owner
R1	*Key member unavailable or sick during critical deadlines*	People	2	4	8	Assign a backup role for every key position (as defined in Team Contract); reallocate tasks early.	Team Lead
R2	Integration with payment gateway sandbox (Stripe/VNPay) takes longer than expected	Technology	3	3	9	Prepare mock donation endpoints and demo wallet as a fallback to ensure donation flow works without third-party API issues.	Developer
R3	Scope creep: feature overload (e.g., Charity Marketplace, Volunteer Task Board) leads to missed deadlines	Estimation	3	4	12	Strictly follow MoSCoW prioritization; prioritize Must-have features first and shift Could-have features to out-of-scope if time is constrained	BA
R4	UI/UX designs from Figma are difficult or time-consuming to	Technology	2	3	6	Designer and Developer review Figma components together early to	Designer

	implement in React/Tailwind CSS					ensure design feasibility and use pre-built Tailwind UI components.	
R5	Inaccurate real-time donation progress updates or transaction inconsistency under concurrent simulated donations	Requirements	2	4	8	Use PostgreSQL ACID transactions to ensure database consistency; conduct thorough unit testing for donation logic.	Developer
R6	Differences in coding styles or Git conflict management issues when multiple members merge code into the main branch.	Organizational	3	2	6	Enforce strict git workflows (protecting the `main` branch, requiring at least one Pull Request review before merging) and use Conventional Commits.	Team Lead
R7	Ambiguity or misunderstandings in feature requirements (e.g., case verification rules or donation progress calculation rules).	Requirements	2	3	6	The Business Analyst clearly documents use cases and reviews them carefully with the team and TA during weekly meetings before coding begins.	BA

6 Tools & Collaboration Setup

One of the main goals of the project is to practise teamwork for professional software development. The use of tools is monitored and graded (teamwork component). The team must use:

- ✔ Moodle – announcements and PA submissions.
- ✔ Communication – Discord
- ✔ Task board – GitHub Projects: Link
- ✔ Git repository – GitHub: Link
  - main branch is protected; changes are merged through pull requests with at least one review
  - README (description, setup), .gitignore; never commit secrets (.env, API keys, passwords)
  - commit messages follow the Conventional Commits style (feat:, fix:, docs:, test:, ...)
- ✔ Weekly status meeting (15 minutes) and weekly status report (template provided), committed to /docs/management/weekly every week from week 2 to week 10.
- ✔ Weekly deadline: 23:30 Tuesday.

Repository structure:

Folder	Content
/src	Source code
/docs/management	Project plan; weekly status reports in /docs/management/weekly
/docs/requirements	Vision, SRS, use cases, mockups

/docs/design	Architecture (C4), ADRs, UML models, UI design
/docs/test	Test plan, test cases, test reports
/docs/ai-usage	AI usage logs
/pa/PA1 … /pa/PA5	One sub-folder per PA submission (PDF + source files)

7 AI Usage Log

[Required in every PA (course GenAI policy). List each significant use of AI tools. Write "No AI tools were used" if none. Undisclosed AI use is treated as an academic integrity violation.]

#	Member	Tool	Task / purpose	How the output was verified and changed
1	Trần Anh Khoa	ChatGPT	Launch task plan, manage project schedule	Discussed the suggestions with the team and adjusted them to fit our project.
2	Lê Nho Duy Phước	Claude	Draft the MoSCoW feature table and out-of-scope list and the C4 System Context diagram for ImpactHub.	Cross-checked every feature, actor and external system against PA0 and the planned tech stack, and confirmed the diagram stays at the C4 Context level. Wording, priorities and relationship labels were adjusted with the team to match PA0.
3	Nguyễn Trần Phương Thuý	ChatGPT	Review the PA1 proposal and plan: the project overview, the WBS and Gantt timeline, effort and cost estimation, and the risk register.	Cross-checked the content against PA0, the PA1 rubric and the Level-3 WBS work packages, and rechecked the person-hours, costs and RE values. Phase-level estimation was replaced with bottom-up estimation and the milestones and WBS layout were refined, while suggestions that expanded the scope or added a separate prototype phase were rejected.
4	Hồng Gia Bảo	Gemini	Draft the PA1 risk register with RE calculations, the technology stack and operating environment, and the human resource	Checked the RE values (P × L), the 367 person-hours and the 29.360.000 VND cost against the PA1 template, Section 4.3 and the team contract, and confirmed the free-tier hosting fits the MVP. Risk owners, mitigations and

			and cost estimate for ImpactHub.	tool choices were adjusted to match the team's roles and setup, while enterprise risk frameworks, Docker/Kubernetes and paid licence costs were rejected.